

15/04/25

MODULE IV

GROUP THEORY

If G is a non-empty set and $*$ is a binary operation, then, $(G, *)$ is called a group if it satisfies the following conditions:

① Closure property: If $a, b \in G$, then $a * b \in G$

② Associative property: If $a, b, c \in G$, then

$$(a * b) * c = a * (b * c)$$

③ Existence of identity element:

There exist an element $e \in G$, such that

$$a * e = e * a = a, \forall a \in G.$$

④ Existence of inverse:

For each $a \in G$, there exists ~~an inverse~~ $a^{-1} \in G$ such that,

$$a^{-1} * a = a * a^{-1} = e$$

Examples:

① $\mathbb{Z}$, the set of integers under addition is a group.

① Closure property

$$a, b \in \mathbb{Z}, \text{ then } a + b \in \mathbb{Z}$$

$\therefore$ satisfies closure property

② Associative property

$$a, b, c \in \mathbb{Z}, \text{ then } (a + b) + c = a + (b + c)$$

$\therefore$ satisfies

③ Identity element

$$a, e \in \mathbb{Z}, \quad a + e = e + a = a$$

$$\text{then } e = 0$$

0 is the identity element.

④ Existence of inverse

$$a + (-a) = e = 0 = (-a) + a$$

$\therefore$ inverse of a is $(-a)$

$\therefore$ It is a group.

- ② $\mathbb{Q}$, set of all rational numbers is a group under addition.
- ③ $\mathbb{R}$, real numbers is a group under addition.
- ④ $\mathbb{C}$, set of all complex no.s under addition is a group.

SEMI GROUP

satisfies property ① and ②.

MONOID

satisfies property ①, ② and ③.

COMMUTATIVE / ABELIAN GROUP

A group which satisfies commutative property also.
i.e., $a * b = b * a$ is known as abelian group.

Examples of not a group

① $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are all not group under the operation multiplication.

→ Identity element — 1.

→ Inverse does not exist for $\mathbb{Z}$ and for 0. Inverse does not exist for $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$.

Q. ① ST the set $\mathbb{Q}^+$, the set of +ve rational no.s form an abelian group under the operation $*$ defined by $a * b = \frac{ab}{2}$, $a, b \in \mathbb{Q}^+$

Ans:

① Closure property

let $a, b \in \mathbb{Q}^+$, then $a * b = \frac{ab}{2} \in \mathbb{Q}^+$

$\therefore$ closure property satisfied

② Associative property

let $a, b, c \in \mathbb{Q}^+$, then $(a * b) * c = \left(\frac{ab}{2}\right) * c$

$$= \frac{abc}{4}$$

$$a * (b * c) = a * \left(\frac{bc}{2}\right)$$

$$= \frac{abc}{4}$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ associative property satisfied

③ Existence of identity

let 'e' be the identity element,

$$a * e = e * a = a$$

$$\text{then, } \frac{ae}{2} = a$$

$$\Rightarrow e = 2, \text{ is the identity}$$

$$e = 2 \in \mathbb{Q}^+$$

$\therefore$ identity element exists

④ Existence of inverse

let a^{-1} be the inverse of a .

$$\text{then, } a * a^{-1} = e$$

$$\frac{aa^{-1}}{2} = \frac{1 \cdot 1}{2} = \frac{1}{2} \text{ (identity element)} \quad \textcircled{1}$$

$$a^{-1} = \frac{4}{a} \in \mathbb{Q}^+ \text{ (identity element)}$$

$\therefore$ inverse exists.

$\therefore$ It is a group.

commutative property

let $a, b \in \mathbb{Q}^+$, then $a * b = \frac{ab}{2}$

$$b * a = \frac{ba}{2}$$

$$= \frac{ab}{2} = a * b$$

$\therefore$ commutative property satisfied

$\therefore$ It is an abelian group.

Q2. ST $(A, \cdot)$ is an abelian group where

$A = \{a \in \mathbb{Q}, a \neq -1\}$ and $a \cdot b = a + b + ab$.

① closure property

let $a, b \in A$, then $a \cdot b = a + b + ab \in A$

$\therefore$ closure property is satisfied

② Associative property

let $a, b, c \in A$, then $a \cdot (b \cdot c) = a \cdot (b + c + bc)$

$$= a + b + c + bc + a(b + c + bc)$$

$$= a + b + c + bc + ab + ac + abc$$

$$(a \cdot b) \cdot c = (a + b + ab) \cdot c$$

$$= a + b + ab + c + ac + bc + abc$$

$$= a \cdot (b \cdot c)$$

$\therefore$ associative property is satisfied

③ Existence of identity element

let 'e' be identity element,

$$\text{then, } a \cdot e = a = e \cdot a$$

$$\Rightarrow a + e + ae = a$$

$$e + ae = 0$$

$$e(1+a) = 0$$

$$\text{Since, } a \neq -1, e = 0 \in A$$

$\therefore 0$, is the identity element

④ Existence of inverse

let a^{-1} be the inverse,

$$a \cdot a^{-1} = e = a^{-1} \cdot a$$

$$\Rightarrow a \cdot a^{-1} = 0$$

$$a + a^{-1} + aa^{-1} = 0$$

$$a^{-1}(1+a) = -a$$

$$a^{-1} = \frac{-a}{1+a} \in A$$

$\therefore a^{-1}$ exist for $(A, \cdot)$

$\therefore$ It is a group

⑤

Commutative property

$$\text{let } a, b \in A, a \cdot b = a + b + ab$$

$$= b + a + ba$$

$$= b \cdot a$$

$\therefore$ commutative property satisfied

$\therefore$ It is an abelian group

Q3. Let $\mathbb{R}^{2 \times 2}$, set of all 2×2 real matrices is a group under matrix addition. Is it an abelian group?

Ans: ① closure property

$$\text{let } A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}, B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

$$\text{then, } A+B = \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

$\therefore$ closure property satisfied

② Associative property

$$\text{let } A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}, B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix}, C = \begin{bmatrix} a_3 & b_3 \\ c_3 & d_3 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$$

$$(A+B)+C = \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{bmatrix} + C$$

$$= \begin{bmatrix} a_1+a_2+a_3 & b_1+b_2+b_3 \\ c_1+c_2+c_3 & d_1+d_2+d_3 \end{bmatrix}$$

$$A+(B+C) = A + \begin{bmatrix} a_2+a_3 & b_2+b_3 \\ c_2+c_3 & d_2+d_3 \end{bmatrix}$$

$$= \begin{bmatrix} a_1+a_2+a_3 & b_1+b_2+b_3 \\ c_1+c_2+c_3 & d_1+d_2+d_3 \end{bmatrix}$$

$$= \underline{\underline{(A+B)+C}}$$

$\therefore$ associative property satisfied

③ Identity element

$$\text{let } E = \begin{bmatrix} m & n \\ 0 & p \end{bmatrix}$$

$$A + E = A$$

$$\begin{bmatrix} a_1 & b_1 \\ c & d_1 \end{bmatrix} + \begin{bmatrix} m & n \\ 0 & p \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ c & d_1 \end{bmatrix}$$

$$\begin{bmatrix} a_1 + m & b_1 + n \\ c + 0 & d_1 + p \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ c & d_1 \end{bmatrix}$$

$$\Rightarrow m = n = 0, p = 0$$

$$E = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\therefore$ Identity element exists.

④ Existence of inverse

$$\text{let } A^{-1} = \begin{bmatrix} a_5 & b_5 \\ c_5 & d_5 \end{bmatrix}$$

$$A + A^{-1} = E$$

$$\begin{bmatrix} a_1 & b_1 \\ c & d_1 \end{bmatrix} + \begin{bmatrix} a_5 & b_5 \\ c_5 & d_5 \end{bmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -a_1 & -b_1 \\ -c & -d_1 \end{bmatrix} = -A$$

$\therefore$ Inverse exists.

$\therefore$ It is a group.

commutative property

$$A+B = \begin{pmatrix} a_1+a_2 & b_1+b_2 \\ c_1+c_2 & d_1+d_2 \end{pmatrix} = \begin{pmatrix} a_2+a_1 & b_2+b_1 \\ c_2+c_1 & d_2+d_1 \end{pmatrix} = B+A$$

$\therefore$ commutative property satisfied.

$\therefore$ It is a commutative group.

Q4. ST $(A, *)$ be a non-abelian group where

$A = \mathbb{R} \times \mathbb{R}$ and for any $(a, b) \in A$,

$$(a, b) * (c, d) = (\underline{ac}, \underline{bc+d})$$

Ans: ① closure property

let $a, b, c, d \in A$,

$$(a, b) * (c, d) = (\underline{ac}, \underline{bc+d}) \in A$$

$\therefore$ closure property satisfied

② Associative property

$$\begin{aligned} [(a, b) * (c, d)] * (e, f) &= [(\underline{ac}, \underline{bc+d})] * (e, f) \\ &= (\underline{ace}, \underline{bce+de+f}) \end{aligned}$$

$$\begin{aligned} (a, b) * [(c, d) * (e, f)] &= (a, b) * (\underline{ce}, \underline{de+f}) \\ &= (\underline{ace}, \underline{bce+de+f}) \end{aligned}$$

$$= [(a, b) * (c, d)] * (e, f)$$

$\therefore$ associative property satisfied

③ Existence of Identity property

let (e_1, e_2) be identity,

$$(a, b) * (e_1, e_2) = (a, b)$$

$$(ae_1, be_2) = (a, b)$$

$$\Rightarrow ae_1 = a$$

$$\underline{e_1 = 1}$$

$$be_2 = b$$

$$\underline{e_2 = 0}$$

$\therefore (1, 0)$ is the identity element

④ Existence of inverse

Let (a^{-1}, b^{-1}) be the inverse

$$(a, b) * (a^{-1}, b^{-1}) = (1, 0)$$

$$(aa^{-1}, ba^{-1} + b^{-1}) = (1, 0)$$

$$\Rightarrow aa^{-1} = 1$$

$$\Rightarrow ba^{-1} + b^{-1} = 0$$

$$\underline{a^{-1} = 1/a}$$

$$\underline{b^{-1} = -b/a}$$

$$\therefore (a^{-1}, b^{-1}) = \left(\frac{1}{a}, -b/a\right)$$

$\therefore$ inverse exists

$\therefore$ It is a group

Commutative property

$$(a, b) * (c, d) = (ac, bc + d)$$

$$(c, d) * (a, b) = (ca, da + b)$$

$$\Rightarrow (a, b) * (c, d) \neq (c, d) * (a, b)$$

$\therefore$ commutative property is not satisfied

$\therefore$ It is a non-abelian group.

hence proved

Q5. M_2 , set of all 2×2 non-singular matrices over $\mathbb{R}$.

$M_2 = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix}, a, b, c, d \in \mathbb{R}, ad - bc \neq 0 \right\}$ is a group

under matrix multiplication?

is it an abelian group?

group (no)

① closure ~~commutative~~ property.

let $A = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, B = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \in M_2$

$a_1 d_1 - b_1 c_1 \neq 0$

$a_2 d_2 - b_2 c_2 \neq 0$.

~~$A+B$~~ $AB = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}$

$$= \begin{pmatrix} a_1 a_2 + b_1 c_2 & a_1 b_2 + b_1 d_2 \\ c_1 a_2 + d_1 c_2 & c_1 b_2 + d_1 d_2 \end{pmatrix}$$

Here $(a_1 a_2 + b_1 c_2)(c_1 b_2 + d_1 d_2) - (a_1 b_2 + b_1 d_2)(c_1 a_2 + d_1 c_2)$

$$= a_1 a_2 c_1 b_2 + a_1 a_2 d_1 d_2 + b_1 b_2 c_1 c_2 + b_1 c_2 d_1 d_2$$

$$- a_1 a_2 b_2 c_1 - a_1 b_2 d_1 c_2 - b_1 d_2 c_1 a_2 - b_1 d_2 d_1 c_2$$

Since, $|A| \neq 0$ and $|B| \neq 0$, $|AB| \neq 0$.

So, it is closed under multiplication.

② Associative property

$$(AB)C = \left[\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \right] \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix} = \begin{pmatrix} a_1 a_2 + b_1 c_2 & a_1 b_2 + b_1 d_2 \\ c_1 a_2 + d_1 c_2 & c_1 b_2 + d_1 d_2 \end{pmatrix} \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix}$$

$$= \begin{pmatrix} a_1 a_2 a_3 + b_1 c_2 a_3 + a_1 b_2 c_3 + b_1 d_2 c_3 & a_1 a_2 b_3 + b_1 c_2 b_3 + a_1 b_2 d_3 + b_1 d_2 d_3 \\ c_1 a_2 a_3 + d_1 c_2 a_3 + c_1 a_2 b_3 + c_1 b_2 d_3 + d_1 c_2 b_3 + d_1 d_2 d_3 & c_1 a_2 b_3 + c_1 b_2 d_3 + d_1 c_2 b_3 + d_1 d_2 d_3 \end{pmatrix}$$

$$A(BC) = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \left[\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \begin{pmatrix} a_3 & b_3 \\ c_3 & d_3 \end{pmatrix} \right] = A \begin{pmatrix} a_2 a_3 + b_2 c_3 & a_2 b_3 + b_2 d_3 \\ c_2 a_3 + d_2 c_3 & c_2 b_3 + d_2 d_3 \end{pmatrix}$$

$$= \begin{pmatrix} a_1 a_2 a_3 + a_1 b_2 c_3 & a_1 a_2 b_3 + a_1 b_2 d_3 + b_1 c_2 a_3 + b_1 d_2 c_3 & b_1 b_2 c_3 + b_1 d_2 d_3 \\ c_1 a_2 a_3 + c_1 b_2 c_3 + d_1 c_2 a_3 + d_1 d_2 c_3 & c_1 a_2 b_3 + c_1 b_2 d_3 + d_1 c_2 b_3 + d_1 d_2 d_3 \end{pmatrix}$$

$$= (AB)C$$

$\therefore$ Associative property satisfied

③ Identity,

$$AE = A \Rightarrow \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a_1 a + b_1 c & a_1 b + b_1 d \\ c_1 a + d_1 c & c_1 b + d_1 d \end{pmatrix} = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}$$

$$a_1 a + b_1 c = a_1, \quad a_1 b + b_1 d = b_1$$

$$c_1 a + d_1 c = c_1, \quad c_1 b + d_1 d = d_1$$

$$\Rightarrow E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

④ Inverse.

$$A \cdot A^{-1} = E \Rightarrow A^{-1} = \frac{1}{|A|} \begin{bmatrix} d_1 & -a_1 \\ -b_1 & a_1 \end{bmatrix} = \frac{1}{a_1 d_1 - b_1 c_1} \begin{bmatrix} d_1 & -a_1 \\ -b_1 & a_1 \end{bmatrix}$$

$\therefore$ It is a group.

Commutative property

$$AB \neq BA$$

$\therefore$ It is not an abelian group.

THEOREM

TH₁ the identity element in a group is unique.

Assume that e_1 and e_2 be the identity elements in a group G .

CASE - I

Consider e_1 is the identity, then

$$e_1 e_2 = e_2 = e_2 e_1 \text{ — ①}$$

CASE - II

consider e_2 is the identity, then

$$e_1 e_2 = e_1 = e_2 e_1 \text{ — ②}$$

From ① and ②,

$$\begin{aligned} e_1 &= e_1 e_2 \\ &= e_2 \end{aligned}$$

$$\text{i.e., } \underline{e_1 = e_2}$$

$\therefore$ identity element in a group is unique.

TH-2

The inverse of each element in a group is unique.

let $a \in G$ and let b , and c are inverse of a .

First, assume that b is the inverse of a .

$$ab = e = ba \text{ — ①}$$

Again, assume c is the inverse of a .

$$ac = e = ca \text{ — ②}$$

$$\begin{aligned} b &= be = b(ac) = (ba)c \\ &= ec \\ &= c \end{aligned}$$

i.e., $\underline{b = c}$

$\therefore$ inverse of each element is unique

TH 3. If $a, b \in G$ and $ba = bc \Rightarrow a = c$
(left-cancellation property)

$ba = bc$

multiplying b^{-1} on both sides;

$b^{-1}ba = b^{-1}bc$

$(b^{-1}b)a = (b^{-1}b)c$

$ea = ec$

$a = c$

Examples:

Q. $G = \{1, +, i, -i\}$, PT this a group under multiplication.

① closure property

	1	+	i	-i
1	1	+	i	-i
+	+	-1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

$\therefore$ closure property satisfies

② Associative property

$a(bc) = (ab)c$

$\therefore$ associative property satisfied

③ Identity element.
Here, 1 is the identity element.
 $\therefore$ identity element exists.

④ Inverse of elements.

Inverse of 1 is 1

" -1 is -1

" i is -i

" -i is i

$\therefore$ inverse exists.

$\therefore$ It is a group.

commutative property

From the table, commutative property also satisfies.

$\therefore$ It is an abelian group.

SUBGROUP

Let G be a group and H is a subset of G ,
then H is a subgroup of G if H itself is a group
under the same operation of G .

Example: $(\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Q}, +)$

ORDER OF A GROUP & ORDER OF AN ELEMENT IN

No. of elements in a group is called order of group.

The order of an element 'a' of a group G is the
smallest +ve integer n such that, $a^n = e$

In the above table, $O(G) = 4$

$$O(1) = 1$$

$$O(i) = 4$$

$$O(-1) = 2$$

$$O(-i) = 4$$

Q. PT 'addition modulo 6' is an abelian group.

$$(Z_6, +_6)$$

Here, $Z_6 = \{0, 1, 2, 3, 4, 5\}$

$+_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

① Closure property

$$a +_6 b \in Z_6$$

$\therefore$ closure property satisfied

② Associative property

$$(a +_6 b) +_6 c = a +_6 (b +_6 c)$$

$\therefore$ associative

③ Identity element

identity element is 0.

④ Inverse element

Inverse of 0 is 0

" 1 is 5

" 2 is 4

" 3 is 3 = (1)0

" 4 is 2 = (1-)0

" 5 is 1

commutative property

$$a \pm b = b \pm a$$

$\therefore$ It is an abelian group.

Q. check $(\mathbb{Z}_6, \times_6)$ is an abelian group.

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

$\times_6$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

① closure property

$$a \times_6 b \in \mathbb{Z}$$

② Associative property

$$(a \times_6 b) \times_6 c = a \times_6 (b \times_6 c)$$

③ Identity

Identity is 1.

$\therefore$ identity exists.

④ Inverse,

Inverse doesn't exist for 0, 2, 3

$\therefore$ inverse does not exist.

$\therefore$ It is not an abelian group.

CYCLIC GROUP

A group G is called a cyclic group if every element of G can be expressed as some power of a particular element, $a \in G$.

That element is called generator.

Example: $G = \{1, -1, i, -i\}$ is a cyclic group.

Here, the generators are i , and $-i$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = -i$$

$$i^4 = 1$$

$$(-i)^1 = -i$$

$$(-i)^2 = -1$$

$$(-i)^3 = +i$$

$$(-i)^4 = 1$$

THEOREM

→ A cyclic group is abelian.

Let $a, b \in G$, then $a = x^m$

$$b = x^n$$

$$\therefore ab = x^m \cdot x^n = x^{m+n}$$

$$= x^{n+m}$$

$$= ba$$

$$\text{i.e., } ab = ba$$

$\therefore$ cyclic group is abelian

THEOREM

→ A group G is abelian iff $(ab)^2 = a^2b^2$, $a, b \in G$.

Let $a, b \in G$,

First, we assume that G is abelian, then

we have to prove $(ab)^2 = a^2b^2$

$$ab = ba$$

$$\begin{aligned}(ab)^2 &= (ab)(ab) \\ &= a(ba)b \\ &= a(ab)b, \text{ since } ab = ba \\ &= (aa)(bb) \\ &= \underline{a^2b^2}\end{aligned}$$

conversely, we assume that $(ab)^2 = a^2b^2$, then we have to prove G is abelian.

$$(ab)^2 = a^2b^2$$

$$(ab)(ab) = a \cdot a \cdot b \cdot b$$

$$a(ba)b = a(ab)b$$

by using left-cancellation property,

$$(ba)b = (ab)b$$

by using right-cancellation law,

$$\underline{ba = ab}$$

$\therefore \underline{G}$ is abelian.

PERMUTATION GROUP

Permutation on a set A is the one-to-one onto functions.

$$A = \{1, 2, 3\}$$

The different permutations are:

$$P_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$P_4 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$P_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$P_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$P_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$P_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

	P_1	P_2	P_3	P_4	P_5	P_6
P_1	P_1	P_2	P_3	P_4	P_5	P_6
P_2	P_2	P_1	P_6	P_5	P_4	P_3
P_3	P_3	P_4	P_5	P_6	P_1	P_2
P_4	P_4	P_3	P_2	P_1	P_6	P_5
P_5	P_5	P_6	P_1	P_2	P_3	P_4
P_6	P_6	P_5	P_4	P_3	P_2	P_1

From table,
it is a group
but not an
abelian group.

It is a non-
abelian group.

16/04/25 PERMUTATION GROUP

The set of all permutation on a non-empty set under the binary operation composition of permutations is called permutation group.

Permutation group is made up of elements which are the elements of the set.

A permutation on a set is the one-to-one onto functions. If $A = \{1, 2, 3\}$, the permutation group is called symmetric group and is denoted by S_3 , the no. of elements in $S_3 = 3! = 6$.

If $A = \{1, 2, 3, \dots, n\}$, this symmetric group is denote by S_n and the no. of elements is $n!$.

Q1. $G = (\mathbb{Z}_5, +_5)$

- Is it a group?
- Is it an abelian group?
- Find the order of each element.
- Is it a cyclic group, if so, find the generators?

$$G = \{0, 1, 2, 3, 4\}$$

τ_5	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

(i) closure property
From table, τ_5 is closed under τ_5 .

Associative property

$$\forall a, b, c \in G, (a+b)+c = a+(b+c)$$

$\therefore$ associativity satisfied.

Identity

From table, 0 is the identity element

Inverse

The inverse of 0 is 0

" 1 is 4

" 2 is 3

" 3 is 2

" 4 is 1

$\therefore$ It is a group.

(ii) Commutative property

$$a \tau_5 b = b \tau_5 a$$

From table, it is commutative

$\therefore$ It is an abelian group.

(iii) $O(a) = 5$ order of each element $a + a + \dots + a = e$
 $O(0) = 1$ $O(2) = 5$
 $O(1) = 5$ $O(3) = 5$
 $O(4) = 5$

Here, $1 + 1 + 1 + 1 + 1 = 0$, $O(1) = 5$

$2 + 2 + 2 + 2 + 2 = 0$, $O(2) = 5$

$3 + 3 + 3 + 3 + 3 = 0$, $O(3) = 5$

$4 + 4 + 4 + 4 + 4 = 0$, $O(4) = 5$

(iv) cyclic group

$G = (\{0, 1, 2, 3, 4\}, +_5)$

consider 3,

$3^0 = 1$	$3^3 = 3 + 3 + 3 = 4$
$3^1 = 3$	$3^4 = 2$
$3^2 = 3 + 3 = 1$	$3^5 = 0$

$\therefore 3$ is a generator.

Consider 1,

$1^0 = 1$	$1^3 = 3$
$1^1 = 1$	$1^4 = 4$
$1^2 = 2$	$1^5 = 0$

$\therefore 1$ is a generator

Consider 2,

$2^0 = 1$	$2^4 = 3$
$2^1 = 2$	$2^5 = 0$
$2^2 = 4$	
$2^3 = 1$	

$\therefore 2$ is a generator (ii)

consider 4,

$4^0 = 1$	$4^4 = 1$
$4^1 = 4$	$4^5 = 0$
$4^2 = 3$	
$4^3 = 2$	

$\therefore 4$ is a generator

$\therefore$ the generators are 1, 2, 3, 4.

$\therefore$ this is a cyclic group.

NOTE generators are those which have gcd as 1 with element $\frac{1}{2}$ divisor.

Q Consider $(\mathbb{Z}_{12}, +_{12})$, find the generators.

$$\mathbb{Z}_{12} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}.$$

$(\mathbb{Z}_n, +_n)$

The generators are 1, 5, 7, 11

(ii) $(\mathbb{Z}_5, +_5)$

$$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$$

The generators are 1, 2, 3 and 4.

Q Find the order of each element of $(\mathbb{Z}_4, +_4)$

$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$

order of each element,

$$o(0) = 1$$

$$1 +_4 1 +_4 1 +_4 1 = 0, \quad o(1) = 4$$

$$2 +_4 2 = 0, \quad o(2) = 2$$

$$3 +_4 3 +_4 3 +_4 3 = 0, \quad o(3) = 4$$

(ii) $(\mathbb{Z}_6, +_6)$, find order each element & generators.

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

order of each element,

$$o(0) = 1$$

$$o(1) = 6$$

$$2 +_6 2 +_6 2 = 0, \quad o(2) = 3$$

$$3 +_6 3 = 0, \quad o(3) = 2$$

$$4 +_6 4 +_6 4 = 0, \quad o(4) = 3$$

$$5 +_6 5 +_6 5 +_6 5 +_6 5 +_6 5 = 0,$$

$$o(5) = 6$$

The generators are 1 and 5

In a group G ,

$$\textcircled{1} (a^{-1})^{-1} = a, \quad a \in G$$

$$\textcircled{2} (ab)^{-1} = b^{-1}a^{-1}, \quad a, b \in G$$

PROOF

$$\textcircled{1} \text{ Let } a \in G, \text{ then } a^{-1} \in G$$

$$\Rightarrow aa^{-1} = e$$

$$\therefore \underline{(a^{-1})^{-1} = a}$$

$$\textcircled{2} \text{ Consider, } (ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1}$$

$$= aea^{-1}$$

$$= aa^{-1}$$

$$= \underline{e}$$

$$\text{Now, } (b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b$$

$$= b^{-1}eb$$

$$= b^{-1}b$$

$$= \underline{e}$$

$$\therefore (ab)(b^{-1}a^{-1}) = e = (b^{-1}a^{-1})(ab)$$

$$\underline{(ab)^{-1} = b^{-1}a^{-1}}$$

Th Let H be a non-empty subset of a group G .
Then H is a sub-group of G iff:

① $a, b \in H \Rightarrow ab \in H$

② $a \in H \Rightarrow a^{-1} \in H$

PROOF

Let H be a subgroup of G , then H is a group under the same operation in G .

Hence, H satisfies all the conditions of a group.

$\therefore$ ① & ② satisfies.

Conversely, H is a non-empty subset which satisfies

① and ②. Then we to prove H is a subgroup.

Now, $a, b, c \in H$, then

associative property satisfies in G ,
so, in H also.

Now, $a \in H \Rightarrow a^{-1} \in H$

then by ① $a a^{-1} \in H$, $H \cap H = H$
 $\Rightarrow e \in H$

$\therefore$ identity element exists.

$\therefore H$ is a subgroup.

THEOREM

Let H and K be two subgroup of G , then
prove that $H \cap K$ is also a subgroup.

Ans Let $a \in (H \cap K)$, $b \in (H \cap K)$

then we have to prove $ab \in (H \cap K)$

Let $a, b \in H$, $a, b \in K$

then $ab \in H$ and $ab \in K$

Since $a \in (H \cap K)$,
 $\Rightarrow a \in H$ and $a \in K$
 $b \in (H \cap K)$
 $\Rightarrow b \in H$ and $b \in K$

$$a \in H, b \in H \Rightarrow ab \in H$$

$$a \in K, b \in K \Rightarrow ab \in K$$

$$\therefore \boxed{ab \in H \cap K}$$

$a \in H \cap K$, then we have to prove $a^{-1} \in H \cap K$

$$a \in H \cap K \Rightarrow a \in H \text{ and } a \in K$$

$$\Rightarrow a^{-1} \in H \text{ and } a^{-1} \in K$$

$$\Rightarrow a^{-1} \in (H \cap K)$$

$\therefore H \cap K$ is a subgroup.

COSETS

Let H be a subgroup of G and $a \in G$, then
 $aH = \{ ah \mid h \in H, a \in G \}$ is called the left-coset
of H in G .

Similarly, $Ha = \{ ha \mid h \in H, a \in G \}$ is called the
right-coset of H in G .

If the operation is addition, the left-coset is

$$a + H$$

and right-coset is $H + a$.

Example: $G = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$

(G is a group under addition.)

Let $H = \{ \dots, -8, -4, 0, 4, 8, \dots \}$, is a subgroup
under addition.

Here, the left coset are:

$$\textcircled{1} \quad 0+H = H$$

$$\textcircled{2} \quad 1+H = \{ \dots, -7, -3, 1, 5, 9, \dots \}$$

$$\textcircled{3} \quad 2+H = \{ \dots, -6, -2, 2, 6, 10, \dots \}$$

$$\textcircled{4} \quad 3+H = \{ \dots, -5, -1, 3, 7, 11, \dots \}$$

$$(0+H) \cup (1+H) \cup (2+H) \cup (3+H) = G$$

RESULT: If H is a subgroup of G , which is finite and $a, b \in G$, then,

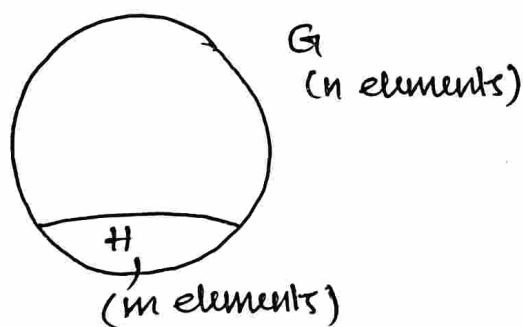
$$|aH| = |H|$$

$$\text{Also, } aH \cap bH = \emptyset$$

THEOREM

Lagrange's Theorem

If G is a finite group of order ' n ' with H is a subgroup of G of order ' m ', then ' m divides n '.



PROOF

Let G be a finite group of order ' n ' and H is a subgroup of G of order ' m '.

Now, the different left-coset of H in G are,

$$a_1H, a_2H, a_3H, \dots, a_kH.$$

$$\text{Then, } G = a_1H \cup a_2H \cup \dots \cup a_kH$$

Also, any two distinct left-cosets of H in G are disjoint.

$$\therefore |G| = |a_1H| + |a_2H| + \dots + |a_kH|$$

$$n = m + m + \dots + m \quad (k \text{ times}) \quad (1)$$

$$n = km \quad (2)$$

$$\Rightarrow m \text{ divides } n \quad (3)$$

i.e., order of subgroup divides order of group.

$$G = (H+0) \cup (H+1) \cup (H+2) \cup \dots \cup (H+(k-1))$$

DEFINITION: If H is a subgroup of G , then the set of cosets of H in G is denoted by G/H .

HOMOMORPHISM & ISOMORPHISM

→ Let $(X, \circ)$ and $(Y, *)$ be any two groups.

Then the mapping $f: (X, \circ) \rightarrow (Y, *)$ is called homomorphism if for any 2 elements $\alpha_1, \alpha_2 \in X$ such that,

$$f(\alpha_1 \circ \alpha_2) = f(\alpha_1) * f(\alpha_2)$$

If the fn. f is one-one and onto, then it is called isomorphism.

Example:

→ Let $(\mathbb{Z}^+, +)$ and $(\mathbb{Z}^+, \times)$ are two groups and mapping is defined by $f(n) = 2^n$. Is it a homomorphism.

Ans:- Let $n, m \in \mathbb{Z}^+$

$$\therefore f(n+m) = 2^{n+m} = 2^n \cdot 2^m = f(n) \times f(m)$$

$\therefore$ these functions form a homomorphism.

Q. 8th root of unity form a group under multiplication.

$$z^4 = 1$$

$$(z^4 - 1) = 0$$

$$z = 1, -1, i, -i$$

Here, $G = \{1, -1, i, -i\}$

(Done previously)

Q2. ST the cube root of unity under multiplication is a group. Is it cyclic and generator are?

$1, \frac{-1+i\sqrt{3}}{2} = \omega, \frac{-1-i\sqrt{3}}{2} = \omega^2$ are the cube roots of 1.
($\omega^2 + \omega + 1 = 0$)

x	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

① closure property

From table, satisfied

② Associativity

$$a \times (b \times c) = (a \times b) \times c$$

satisfied

③ 1 is identity element

④ Inverse of 1 = 1

$$\omega = \omega^2$$

$$\omega^2 = \omega$$

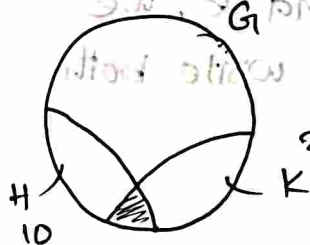
$\therefore$ It is a group.

Here ω and ω^2 are generators.

$\therefore$ it is a cyclic group.

trivial subgroup
(group and
identity
element set)

Q3. If H and K are subgroup of G and e is the identity,
PT 'if $|H|=10$ and $|K|=21$, then $|H \cap K| = \{e\}$ '.



Here, $H \cap K$ is also a subgroup.

Let $|H \cap K| = n$.

$\therefore n$ divides 10 and n divides 21.

$$\therefore \underline{n = 1}$$

$$\therefore \underline{H \cap K = \{e\}}$$

[For any group G , it have 2 trivial
sub-group ($G, \{e\}$) and all others
are non-trivial subgroup]

Hence, proved